Modern Statistical Models

1. Maximum Likelihood: Asymptotics

▶ **Setup** $X_1,\dots,X_n \overset{iid}{\sim} f(x;\boldsymbol\theta_0)$, $\boldsymbol\theta_0\in\Theta\subset\mathbb{R}^p$. Log-likelihood and MLE:

$$\ell_n(\boldsymbol\theta)=\sum_{i=1}^n \log f(X_i;\boldsymbol\theta),\qquad \hat{\boldsymbol\theta}_n=\arg\max_{\boldsymbol\theta\in\Theta}\ell_n(\boldsymbol\theta).$$

▶ **Fisher information** With score $S(x;\boldsymbol\theta)=\partial_{\boldsymbol\theta}\log f(x;\boldsymbol\theta)$,

$$I(\boldsymbol\theta)=\mathbb{E}[S\,S^{\top}]=-\mathbb{E}\Big[\frac{\partial^2}{\partial\boldsymbol\theta\,\partial\boldsymbol\theta^{\top}}\log f\Big]=\mathrm{Var}\,(S(X;\boldsymbol\theta)).$$

▶ **Limit theory** Under regularity conditions, as $n\to\infty$:

$$\hat{\boldsymbol\theta}_n\xrightarrow{P}\boldsymbol\theta_0,\qquad \sqrt{n}\,(\hat{\boldsymbol\theta}_n-\boldsymbol\theta_0)\xrightarrow{d}N(\mathbf{0},\ I^{-1}(\boldsymbol\theta_0)).$$

For large n (finite-sample working approximation):

$$\hat{\boldsymbol\theta}_n\ \dot\sim\ N\Big(\boldsymbol\theta_0,\ \tfrac{1}{n}I^{-1}(\hat{\boldsymbol\theta}_n)\Big),\qquad \hat V:=\tfrac{1}{n}I^{-1}(\hat{\boldsymbol\theta}_n).$$

Here $\frac{1}{n}I^{-1}$ is the inverse information of the *total* sample; $\hat V$ is the estimated covariance used in Wald inference, which is the output of R covariance.

▶ **Information criteria** For model comparison (smaller is better), with $\log L$ the maximized log-likelihood, k parameters, n observations:

$$\mathrm{AIC}=-2\log L+2k,\qquad \mathrm{BIC}=-2\log L+k\log n.$$

▶ **Wald statistics** *Single component* $H_0:\theta_j=\theta^*$:

$$Z=\frac{\hat\theta_j-\theta^*}{\sqrt{\hat V_{jj}}}\xrightarrow{d}N(0,1).$$

Linear restriction $H_0:R\boldsymbol\theta=\boldsymbol r$, rank $R=q$:

$$W=(R\hat{\boldsymbol\theta}-\boldsymbol r)^{\top}(R\hat VR^{\top})^{-1}(R\hat{\boldsymbol\theta}-\boldsymbol r)\xrightarrow{d}\chi^2_q.$$

▶ **Delta method** For $g:\mathbb{R}^k\to\mathbb{R}^m$ with Jacobian $D=\partial g/\partial\boldsymbol\theta^{\top}$,

$$g(\hat{\boldsymbol\theta})\approx g(\boldsymbol\theta_0)+D(\hat{\boldsymbol\theta}-\boldsymbol\theta_0),\qquad g(\hat{\boldsymbol\theta})\ \dot\sim\ N\Big(g(\boldsymbol\theta_0),\ \tfrac{1}{n}D\,I^{-1}(\hat{\boldsymbol\theta}_n)\,D^{\top}\Big).$$

2. Gaussian conditioning

If $\binom{X_1}{X_2}\sim N\Big(\binom{\boldsymbol\mu_1}{\boldsymbol\mu_2},\ \begin{pmatrix}\Sigma_{11}&\Sigma_{12}\\ \Sigma_{21}&\Sigma_{22}\end{pmatrix}\Big)$ then

$$X_1\mid X_2\sim N(\boldsymbol\mu_1+\Sigma_{12}\Sigma_{22}^{-1}(X_2-\boldsymbol\mu_2),\ \Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}).$$

3. Likelihood Ratio Test

Test $H_0:\boldsymbol\theta\in\Theta_0$ vs. $H_1:\boldsymbol\theta\in\Theta\setminus\Theta_0$, where Θ_0 imposes r independent restrictions, with *iid* samples $X_1,\cdots,X_n$. With

$$-2\log\Lambda=2\Big(\sup_{\boldsymbol\theta\in\Theta}\ell_n(\boldsymbol\theta)-\sup_{\boldsymbol\theta\in\Theta_0}\ell_n(\boldsymbol\theta)\Big),$$

under H_0 and regularity conditions $-2\log\Lambda\xrightarrow{d}\chi^2_r$.

4. Classical Linear Model

Model $Y=X\boldsymbol\beta+\boldsymbol\varepsilon$, $\boldsymbol\varepsilon\sim N(\mathbf{0},\sigma^2I_n)$, $X\in\mathbb{R}^{n\times p}$, rank $X=p$.

▶ **OLS & distributions**

$$\hat{\boldsymbol\beta}=(X^{\top}X)^{-1}X^{\top}Y\sim N(\boldsymbol\beta,\ \sigma^2(X^{\top}X)^{-1}),$$

$$\mathrm{RSS}=\|Y-X\hat{\boldsymbol\beta}\|^2,\quad \hat\sigma^2=\frac{\mathrm{RSS}}{n-p},\quad \frac{\mathrm{RSS}}{\sigma^2}\sim\chi^2_{n-p},$$

with $\hat{\boldsymbol\beta}\perp\hat\sigma^2$.

▶ ***t*-test for $\mathbf{c}^{\top}\boldsymbol\beta=0$** For a single contrast vector $\boldsymbol c\in\mathbb{R}^p$:

$$t=\frac{\mathbf{c}^{\top}\hat{\boldsymbol\beta}}{\sqrt{\hat\sigma^2\,\mathbf{c}^{\top}(X^{\top}X)^{-1}\mathbf{c}}}\ \sim\ t_{n-p}\quad\text{under }H_0,$$

CI: $\mathbf{c}^{\top}\hat{\boldsymbol\beta}\pm t_{1-\alpha/2,\,n-p}\sqrt{\hat\sigma^2\,\mathbf{c}^{\top}(X^{\top}X)^{-1}\mathbf{c}}$.

▶ ***F*-test for $C\boldsymbol\beta=0$** $C\in\mathbb{R}^{q\times p}$, rank $C=q$:

$$F=\frac{(C\hat{\boldsymbol\beta})^{\top}(C(X^{\top}X)^{-1}C^{\top})^{-1}(C\hat{\boldsymbol\beta})/q}{\hat\sigma^2}\ \sim\ F_{q,\,n-p}\quad\text{under }H_0.$$

Equivalent reduced-vs-full form (let RSS_0 be the fit under H_0):

$$F=\frac{(\mathrm{RSS}_0-\mathrm{RSS})/q}{\mathrm{RSS}/(n-p)}.$$

▶ **Frisch–Waugh–Lovell (FWL)** Partition $X=[\,X_A\ \ X_B\,]$, $\boldsymbol\beta=(\boldsymbol\beta_A^{\top},\boldsymbol\beta_B^{\top})^{\top}$, so $Y=X_A\boldsymbol\beta_A+X_B\boldsymbol\beta_B+\boldsymbol\varepsilon$. Let

$$P_{X_A}=X_A(X_A^{\top}X_A)^{-1}X_A^{\top},\qquad M_{X_A}=I-P_{X_A}.$$

FWL theorem. The full-model OLS estimate $\hat{\boldsymbol\beta}_B$ equals the OLS estimate from regressing the *residualized* response on the *residualized* regressors $M_{X_A}X_B$:

$$\hat{\boldsymbol\beta}_B=(X_B^{\top}M_{X_A}X_B)^{-1}X_B^{\top}M_{X_A}Y,$$

and the full-model residuals coincide with those of this partialled-out regression. So a coefficient is unchanged by first projecting out the other regressors from both sides ("partialling out").

Projection decomposition. The hat matrix splits orthogonally as

$$P_X=P_{X_A}+M_{X_A}X_B(X_B^{\top}M_{X_A}X_B)^{-1}X_B^{\top}M_{X_A}=P_{X_A}+P_{M_{X_A}X_B},$$

i.e. the column space of X is $\mathrm{col}(X_A)\oplus\mathrm{col}(M_{X_A}X_B)$, two orthogonal pieces. The two projectors are orthogonal ($P_{X_A}P_{M_{X_A}X_B}=0$), which is exactly why $\hat{\boldsymbol\beta}_B$ depends only on the part of X_B orthogonal to X_A.

▶ **Heteroscedastic model(Aitken)** Model $Y=X\boldsymbol\beta+\boldsymbol\varepsilon$, $\boldsymbol\varepsilon\sim(\mathbf{0},\ \sigma^2V)$ with $V\succ0$ *known*. The best linear unbiased estimator is the GLS estimator

$$\hat{\boldsymbol\beta}_{\mathrm{GLS}}=(X^{\top}V^{-1}X)^{-1}X^{\top}V^{-1}Y,\qquad \mathrm{Var}(\hat{\boldsymbol\beta}_{\mathrm{GLS}})=\sigma^2(X^{\top}V^{-1}X)^{-1}.$$

Pure heteroscedasticity $V=\mathrm{diag}(v_1,\dots,v_n)$ gives weighted LS with weights $w_i=1/v_i$.

▶ **REML (variance estimation)** For $Y=X\boldsymbol\beta+\boldsymbol\varepsilon$, $\boldsymbol\varepsilon\sim N(\mathbf{0},\Sigma)$, $r=\mathrm{rank}\,X$. Choose linearly independent $\boldsymbol a_1,\dots,\boldsymbol a_{n-r}$ with $\boldsymbol a_i^{\top}X=\mathbf{0}$; set $A=[\boldsymbol a_1,\dots,\boldsymbol a_{n-r}]^{\top}$, rank $A=n-r$. Then error contrasts

$$AY=A\boldsymbol\varepsilon\sim N(\mathbf{0},\ A\Sigma A^{\top})$$

are free of $\boldsymbol\beta$. Maximize their (restricted) log-likelihood

$$\ell(\Sigma)=-\tfrac{n-r}{2}\log(2\pi)-\tfrac12\log|A\Sigma A^{\top}|-\tfrac12(AY)^{\top}(A\Sigma A^{\top})^{-1}(AY),$$

$\hat\Sigma_{\mathrm{REML}}=\arg\max_\Sigma\ell(\Sigma)$. The value is invariant to the choice of A; REML removes the downward bias of ML variance estimates.

5. Generalized Linear Models

▶ **Bernoulli GLM — Wald CIs** $Y_i\overset{ind}{\sim}\mathrm{Bernoulli}(\pi_i)$, link $g(\pi_i)=\boldsymbol x_i^{\top}\boldsymbol\beta$. With $\hat{\boldsymbol\beta}$ and $\hat V=I^{-1}(\hat{\boldsymbol\beta})$ (inverse Fisher information):

$$\hat\beta_j\pm1.96\sqrt{\hat V_{jj}},\qquad \boldsymbol x^{\top}\hat{\boldsymbol\beta}\pm1.96\sqrt{\boldsymbol x^{\top}\hat V\boldsymbol x}\quad\text{(linear predictor)}.$$

Transform endpoints through g^{-1} for a CI on π.

▶ **Deviance** In R the deviance residual uses

$$D=2\{\ell(\text{saturated})-\ell(\text{fitted})\}.$$

For Bernoulli data the saturated log-likelihood term is 0.
Here, saturated model means that the model parameter λ_i (Poisson case) or π_i(Bernoulli case) is estimated with y_i.

▶ **Poisson GLM residuals** Fitted means $\hat\lambda_i$. **Deviance residual**

$$d_i=\mathrm{sign}(y_i-\hat\lambda_i)\sqrt{2[y_i\log(y_i/\hat\lambda_i)-(y_i-\hat\lambda_i)]},$$

and the LR (deviance) statistic is $\sum_{i=1}^n d_i^2$. **Pearson residual**

$$r_i=\frac{y_i-\hat\lambda_i}{\sqrt{\hat\lambda_i}},\qquad \chi^2=\sum_{i=1}^n r_i^2\ \sim\ \chi^2_{n-p}\quad\text{if the model is correct.}$$

▶ **Overdispersion & quasi-likelihood** If $\mathrm{Var}(Y)\gg\mathbb{E}(Y)$, estimate the dispersion by

$$\hat\phi=\frac{1}{n-p}\sum_{i=1}^n r_i^2\ (\gg1),\qquad \mathrm{Var}(Y)=\hat\phi\,\mathbb{E}(Y).$$

Quasi-likelihood: the point estimate $\hat{\boldsymbol\beta}$ is *unchanged*; the covariance (inverse Fisher information) is scaled by $\hat\phi$, so every standard error is multiplied by $\sqrt{\hat\phi}$.

▶ **Log-normal facts & Poisson–log-normal** If $\log v\sim N(\mu,\sigma^2)$:

$$\mathbb{E}v=e^{\mu+\sigma^2/2},\qquad \mathrm{Var}\,v=e^{2\mu+2\sigma^2}-e^{2\mu+\sigma^2}.$$

Mixture: $\log v\sim N(\mu,\sigma^2)$, $u\mid v\sim\mathrm{Poisson}(v)$. Then

$$\mathbb{E}u=\mathbb{E}[\mathbb{E}(u\mid v)]=\mathbb{E}v=e^{\mu+\sigma^2/2},$$

$$\mathrm{Var}\,u=\mathbb{E}[\mathrm{Var}(u\mid v)]+\mathrm{Var}[\mathbb{E}(u\mid v)]=\mathbb{E}v+\mathrm{Var}\,v=\mathbb{E}u+(e^{\sigma^2}-1)(\mathbb{E}u)^2.$$

The extra term $(e^{\sigma^2}-1)(\mathbb{E}u)^2>0$ is the overdispersion beyond Poisson.

▶ **GLM with random effects** $\ell_i\overset{iid}{\sim}N(0,\sigma^2)$, $\log\lambda_i=\boldsymbol x_i^{\top}\boldsymbol\beta+\ell_i$, $Y_i\mid\lambda_i\sim\mathrm{Poisson}(\lambda_i)$. Marginal pmf (integrate out the log-normal λ_i):

$$P(Y_i=y)=\int_0^\infty\frac{e^{-\lambda}\lambda^y}{y!}\cdot\frac{1}{\lambda\sqrt{2\pi\sigma^2}}\exp\Big\{-\frac{(\log\lambda-\boldsymbol x_i^{\top}\boldsymbol\beta)^2}{2\sigma^2}\Big\}\,d\lambda.$$

This is one mechanism producing the overdispersion above.

6. Linear Mixed-Effects Models

▶ **General LMM, BLUP & EBLUP**

$$Y=X\boldsymbol\beta+Zu+e,\quad u\sim\mathcal{N}(\mathbf{0},G),\ e\sim\mathcal{N}(\mathbf{0},R),\ \mathrm{Cov}(u,e)=0,$$

$$\Sigma:=\mathrm{Var}(Y)=ZGZ^{\top}+R.$$

From the distribution of error, we have

$$\binom{u}{Y}\sim\mathcal{N}\left(\binom{0}{X\beta},\begin{pmatrix}G&GZ^{\top}\\ ZG^{\top}&\Sigma\end{pmatrix}\right),$$

from the standard result of Gaussian conditioning, random effects u can be estimated with

$$\hat u=\mathbb{E}(u\mid Y)=GZ^{\top}\Sigma^{-1}(Y-X\beta).$$

In practice, we usually replace by its GLS estimator $\hat\beta$, yielding the EBLUP(E stands for empirical):

$$\hat u=GZ^{\top}\Sigma^{-1}(Y-X\hat\beta).$$

Below shows a few of the widely used linear mixed models:

▶ **Case I — Subsampling (nested) design** $y_{ijk}=\mu_i+u_{ij}+e_{ijk}$, $u_{ij}\sim N(0,\sigma_u^2)$, $e_{ijk}\sim N(0,\sigma_e^2)$ indep.; $i=1{:}a$, $j=1{:}b$, $k=1{:}n$.

Source	df	SS	EMS
Treatments i	$a-1$	$bn\sum_i(\bar y_{i\cdot\cdot}-\bar y_{\cdot\cdot\cdot})^2$	$\sigma_e^2+n\sigma_u^2+\frac{bn}{a-1}\sum_i(\mu_i-\bar\mu)^2$
Units(i), $j(i)$	$a(b-1)$	$n\sum_{i,j}(\bar y_{ij\cdot}-\bar y_{i\cdot\cdot})^2$	$\sigma_e^2+n\sigma_u^2$
Error	$ab(n-1)$	$\sum_{i,j,k}(y_{ijk}-\bar y_{ij\cdot})^2$	σ_e^2
Total	$abn-1$	$\sum(y_{ijk}-\bar y_{\cdot\cdot\cdot})^2$	

Tests & inference.

- $H_0:\mu_1=\cdots=\mu_a$: $F_A=\dfrac{MS_A}{MS_{U(A)}}\sim F_{a-1,\,a(b-1)}$.
- $H_0:\sigma_u^2{=}0$ vs. $\sigma_u^2{>}0$: $F_U=\dfrac{MS_{U(A)}}{MS_E}\sim F_{a(b-1),\,ab(n-1)}$.
- Contrast $L=\sum_i c_i\mu_i$ ($\sum_i c_i=0$), $\hat L=\sum_i c_i\bar y_{i\cdot\cdot}$:

$$\mathrm{Var}(\hat L)=\frac{\sigma_e^2+n\sigma_u^2}{bn}\sum_i c_i^2,\qquad SE(\hat L)=\sqrt{\frac{MS_{U(A)}}{bn}\textstyle\sum_i c_i^2},$$

$$t=\frac{\hat L-L_0}{SE(\hat L)}\sim t_{a(b-1)},\qquad \hat L\pm t_{1-\alpha/2,\,a(b-1)}SE(\hat L).$$

▶ **Case II — Split-plot design** $y_{ijk}=\mu_{ij}+b_k+w_{ik}+e_{ijk}$. Whole-plot factor A: $i=1{:}a$; subplot factor B: $j=1{:}c$; block $k=1{:}r$. Random: $b_k\sim N(0,\sigma_b^2)$, $w_{ik}\sim N(0,\sigma_w^2)$, $e_{ijk}\sim N(0,\sigma_e^2)$ indep. Marginal means $\mu_{i\cdot}=\frac1c\sum_j\mu_{ij}$, $\mu_{\cdot j}=\frac1a\sum_i\mu_{ij}$, $\mu_{\cdot\cdot}=\frac{1}{ac}\sum_{i,j}\mu_{ij}$, interaction $\delta_{ij}=\mu_{ij}-\mu_{i\cdot}-\mu_{\cdot j}+\mu_{\cdot\cdot}$.

Source	df	SS	EMS
Blocks	$r-1$	$ac\sum_k(\bar y_{\cdot\cdot k}-\bar y_{\cdot\cdot\cdot})^2$	$\sigma_e^2+c\sigma_w^2+ac\sigma_b^2$
Whole-plot A	$a-1$	$cr\sum_i(\bar y_{i\cdot\cdot}-\bar y_{\cdot\cdot\cdot})^2$	$\sigma_e^2+c\sigma_w^2+\frac{cr}{a-1}\sum_i(\mu_{i\cdot}-\mu_{\cdot\cdot})^2$
W. error $E(A)$=Blk×A	$(r-1)(a-1)$	$c\sum_{i,k}(\bar y_{i\cdot k}-\bar y_{i\cdot\cdot}-\bar y_{\cdot\cdot k}+\bar y_{\cdot\cdot\cdot})^2$	$\sigma_e^2+c\sigma_w^2$
Subplot B	$c-1$	$ar\sum_j(\bar y_{\cdot j\cdot}-\bar y_{\cdot\cdot\cdot})^2$	$\sigma_e^2+\frac{ar}{c-1}\sum_j(\mu_{\cdot j}-\mu_{\cdot\cdot})^2$
$A\times B$	$(a-1)(c-1)$	$r\sum_{i,j}(\bar y_{ij\cdot}-\bar y_{i\cdot\cdot}-\bar y_{\cdot j\cdot}+\bar y_{\cdot\cdot\cdot})^2$	$\sigma_e^2+\frac{r}{(a-1)(c-1)}\sum_{i,j}\delta_{ij}^2$
S. error $E(B)$	$a(r-1)(c-1)$	$\sum_{i,j,k}(y_{ijk}-\bar y_{ij\cdot}-\bar y_{i\cdot k}+\bar y_{i\cdot\cdot})^2$	σ_e^2
Total	$acr-1$	$\sum(y_{ijk}-\bar y_{\cdot\cdot\cdot})^2$	

***F*-tests** (each $MS=SS/\mathrm{df}$):

- A main effect: $F_A = \dfrac{MS_A}{MS_{E(A)}} \sim F_{a-1,\,(r-1)(a-1)}$, $H_{0A}: \mu_{1\cdot\cdot} = \cdots = \mu_{a\cdot\cdot}$.
- B main effect: $F_B = \dfrac{MS_B}{MS_{E(B)}} \sim F_{c-1,\,a(r-1)(c-1)}$, $H_{0B}: \mu_{\cdot 1} = \cdots = \mu_{\cdot c}$.
- Interaction: $F_{AB} = \dfrac{MS_{AB}}{MS_{E(B)}} \sim F_{(a-1)(c-1),\,a(r-1)(c-1)}$, $H_{0AB}: \delta_{ij}=0\ \forall i,j$.

Contrasts.

1. Subplots within one whole plot, $L = \mu_{ij_1} - \mu_{ij_2}$:

$$\widehat{\mathrm{Var}}(\hat L) = \tfrac{2}{r} MS_{E(B)}, \quad t = \frac{\bar y_{ij_1\cdot} - \bar y_{ij_2\cdot}}{\sqrt{2MS_{E(B)}/r}} \sim t_{a(r-1)(c-1)}.$$

2. Between A levels, $L = \mu_{i_1\cdot} - \mu_{i_2\cdot}$:

$$SE(\hat L) = \sqrt{\frac{2MS_{E(A)}}{rc}}, \quad t = \frac{\bar y_{i_1\cdot\cdot} - \bar y_{i_2\cdot\cdot}}{\sqrt{2MS_{E(A)}/(rc)}} \sim t_{(r-1)(a-1)}.$$

3. Between B levels (marginal), $L = \mu_{\cdot j_1} - \mu_{\cdot j_2}$:

$$SE(\hat L) = \sqrt{\frac{2MS_{E(B)}}{ar}}, \quad t = \frac{\bar y_{\cdot j_1\cdot} - \bar y_{\cdot j_2\cdot}}{\sqrt{2MS_{E(B)}/(ar)}} \sim t_{a(r-1)(c-1)}.$$

▶ Case III — Two-factor model with random block (repeated measures)

$$y_{ijk} = \mu_{ij} + l_{ik} + e_{ijk}, \qquad \mu_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij},$$

$l_{ik} \sim N(0,\sigma_l^2)$ random, $e_{ijk} \sim N(0,\sigma_e^2)$ indep.; balanced $i = 1{:}a$ (factor A), $j = 1{:}b$ (factor B), $k = 1{:}n$ (subjects/blocks nested in A). Subject (i,k) is the whole-plot unit; A is between-subject, B is within-subject. Constraints $\sum_i \alpha_i = \sum_j \beta_j = \sum_i (\alpha\beta)_{ij} = \sum_j (\alpha\beta)_{ij} = 0$.

Source	df	SS	EMS
A	$a-1$	$bn\sum_i(\bar y_{i\cdot\cdot} - \bar y_{\cdot\cdot\cdot})^2$	$\sigma_e^2 + b\sigma_l^2 + \frac{bn}{a-1}\sum_i \alpha_i^2$
Subj(A)	$a(n-1)$	$b\sum_{i,k}(\bar y_{i\cdot k} - \bar y_{i\cdot\cdot})^2$	$\sigma_e^2 + b\sigma_l^2$
B	$b-1$	$an\sum_j(\bar y_{\cdot j\cdot} - \bar y_{\cdot\cdot\cdot})^2$	$\sigma_e^2 + \frac{an}{b-1}\sum_j \beta_j^2$
$A\times B$	$(a-1)(b-1)$	$n\sum_{i,j}(\bar y_{ij\cdot} - \bar y_{i\cdot\cdot} - \bar y_{\cdot j\cdot} + \bar y_{\cdot\cdot\cdot})^2$	$\sigma_e^2 + \frac{n\sum_{i,j}(\alpha\beta)_{ij}^2}{(a-1)(b-1)}$
Error $B\times$Subj(A)	$a(b-1)(n-1)$	$\sum_{i,j,k}(y_{ijk} - \bar y_{ij\cdot} - \bar y_{i\cdot k} + \bar y_{i\cdot\cdot})^2$	σ_e^2
Total	$abn-1$	$\sum(y_{ijk} - \bar y_{\cdot\cdot\cdot})^2$	

Tests & inference (note the different error terms):

- A (between-subject): $F_A = \dfrac{MS_A}{MS_{\mathrm{Subj}(A)}} \sim F_{a-1,\,a(n-1)}$.
- B (within-subject): $F_B = \dfrac{MS_B}{MS_E} \sim F_{b-1,\,a(b-1)(n-1)}$.
- $A\times B$: $F_{AB} = \dfrac{MS_{AB}}{MS_E} \sim F_{(a-1)(b-1),\,a(b-1)(n-1)}$.
- Within-subject contrast $L = \sum_j c_j\mu_{ij}$ ($\sum_j c_j = 0$): use MS_E, $SE(\hat L) = \sqrt{\frac{MS_E}{n}\sum_j c_j^2}$, $t \sim t_{a(b-1)(n-1)}$.
- Between-subject contrast $L = \sum_i c_i\mu_{i\cdot}$: use $MS_{\mathrm{Subj}(A)}$, $SE(\hat L) = \sqrt{\frac{MS_{\mathrm{Subj}(A)}}{bn}\sum_i c_i^2}$, $t \sim t_{a(n-1)}$.
- Variance component: $\hat\sigma_e^2 = MS_E$, $\hat\sigma_l^2 = \dfrac{MS_{\mathrm{Subj}(A)} - MS_E}{b}$.

▶ Case IV — Split-split-plot design Balanced split-split-plot with one observation per sub-subplot: $y_{ijm\ell} = \mu_{ijm} + \rho_\ell + w_{i\ell} + s_{ij\ell} + e_{ijm\ell}$. Whole-plot factor A: $i = 1{:}a$; subplot factor B: $j = 1{:}b$; sub-subplot factor C: $m = 1{:}c$; block $\ell = 1{:}r$. Random effects $\rho_\ell \sim N(0,\sigma_\rho^2)$, $w_{i\ell} \sim N(0,\sigma_w^2)$, $s_{ij\ell} \sim N(0,\sigma_s^2)$, $e_{ijm\ell} \sim N(0,\sigma_e^2)$ indep.; $w_{i\ell}$, $s_{ij\ell}$, $e_{ijm\ell}$ are the whole-plot, subplot, and sub-subplot errors. Marginal means are defined in the usual way, e.g. $\mu_{i\cdot\cdot} = b^{-1}c^{-1}\sum_{j,m}\mu_{ijm}$, $\mu_{\cdot j\cdot} = a^{-1}c^{-1}\sum_{i,m}\mu_{ijm}$, $\mu_{\cdot\cdot m} = a^{-1}b^{-1}\sum_{i,j}\mu_{ijm}$, $\mu_{\cdot\cdot\cdot} = a^{-1}b^{-1}c^{-1}\sum_{i,j,m}\mu_{ijm}$. Let $\delta_{ij}^{AB} = \mu_{ij\cdot} - \mu_{i\cdot\cdot} - \mu_{\cdot j\cdot} + \mu_{\cdot\cdot\cdot}$, $\delta_{im}^{AC} = \mu_{i\cdot m} - \mu_{i\cdot\cdot} - \mu_{\cdot\cdot m} + \mu_{\cdot\cdot\cdot}$, $\delta_{jm}^{BC} = \mu_{\cdot jm} - \mu_{\cdot j\cdot} - \mu_{\cdot\cdot m} + \mu_{\cdot\cdot\cdot}$, and $\delta_{ijm}^{ABC} = \mu_{ijm} - \mu_{ij\cdot} - \mu_{i\cdot m} - \mu_{\cdot jm} + \mu_{i\cdot\cdot} + \mu_{\cdot j\cdot} + \mu_{\cdot\cdot m} - \mu_{\cdot\cdot\cdot}$.

source-df-EMS

Source	df	EMS
Blocks	$r-1$	$\sigma_e^2 + c\sigma_s^2 + bc\sigma_w^2 + abc\sigma_\rho^2$
Whole-plot A	$a-1$	$\sigma_e^2 + c\sigma_s^2 + bc\sigma_w^2 + \frac{bcr}{a-1}\sum_i(\mu_{i\cdot\cdot} - \mu_{\cdot\cdot\cdot})^2$
W. error $E(A)$=Blk$\times A$	$(r-1)(a-1)$	$\sigma_e^2 + c\sigma_s^2 + bc\sigma_w^2$
Subplot B	$b-1$	$\sigma_e^2 + c\sigma_s^2 + \frac{acr}{b-1}\sum_j(\mu_{\cdot j\cdot} - \mu_{\cdot\cdot\cdot})^2$
$A\times B$	$(a-1)(b-1)$	$\sigma_e^2 + c\sigma_s^2 + \frac{cr}{(a-1)(b-1)}\sum_{i,j}(\delta_{ij}^{AB})^2$
S. error $E(B)$=Blk$\times A\times B$	$a(r-1)(b-1)$	$\sigma_e^2 + c\sigma_s^2$
Sub-subplot C	$c-1$	$\sigma_e^2 + \frac{abr}{c-1}\sum_m(\mu_{\cdot\cdot m} - \mu_{\cdot\cdot\cdot})^2$
$A\times C$	$(a-1)(c-1)$	$\sigma_e^2 + \frac{br}{(a-1)(c-1)}\sum_{i,m}(\delta_{im}^{AC})^2$
$B\times C$	$(b-1)(c-1)$	$\sigma_e^2 + \frac{ar}{(b-1)(c-1)}\sum_{j,m}(\delta_{jm}^{BC})^2$
$A\times B\times C$	$(a-1)(b-1)(c-1)$	$\sigma_e^2 + \frac{r}{(a-1)(b-1)(c-1)}\sum_{i,j,m}(\delta_{ijm}^{ABC})^2$
SSP error $E(C)$	$ab(r-1)(c-1)$	σ_e^2
Total	$abcr-1$	

source-SS

Source	SS
Blocks	$abc\sum_\ell(\bar y_{\cdot\cdot\cdot\ell} - \bar y_{\cdot\cdot\cdot\cdot})^2$
Whole-plot A	$bcr\sum_i(\bar y_{i\cdot\cdot\cdot} - \bar y_{\cdot\cdot\cdot\cdot})^2$
W. error $E(A)$=Blk$\times A$	$bc\sum_{i,\ell}(\bar y_{i\cdot\cdot\ell} - \bar y_{i\cdot\cdot\cdot} - \bar y_{\cdot\cdot\cdot\ell} + \bar y_{\cdot\cdot\cdot\cdot})^2$
Subplot B	$acr\sum_j(\bar y_{\cdot j\cdot\cdot} - \bar y_{\cdot\cdot\cdot\cdot})^2$
$A\times B$	$cr\sum_{i,j}(\bar y_{ij\cdot\cdot} - \bar y_{i\cdot\cdot\cdot} - \bar y_{\cdot j\cdot\cdot} + \bar y_{\cdot\cdot\cdot\cdot})^2$
S. error $E(B)$=Blk$\times A\times B$	$c\sum_{i,j,\ell}(\bar y_{ij\cdot\ell} - \bar y_{ij\cdot\cdot} - \bar y_{i\cdot\cdot\ell} + \bar y_{i\cdot\cdot\cdot})^2$
Sub-subplot C	$abr\sum_m(\bar y_{\cdot\cdot m\cdot} - \bar y_{\cdot\cdot\cdot\cdot})^2$
$A\times C$	$br\sum_{i,m}(\bar y_{i\cdot m\cdot} - \bar y_{i\cdot\cdot\cdot} - \bar y_{\cdot\cdot m\cdot} + \bar y_{\cdot\cdot\cdot\cdot})^2$
$B\times C$	$ar\sum_{j,m}(\bar y_{\cdot jm\cdot} - \bar y_{\cdot j\cdot\cdot} - \bar y_{\cdot\cdot m\cdot} + \bar y_{\cdot\cdot\cdot\cdot})^2$
$A\times B\times C$	$r\sum_{i,j,m}(\bar y_{ijm\cdot} - \bar y_{ij\cdot\cdot} - \bar y_{i\cdot m\cdot} - \bar y_{\cdot jm\cdot} + \bar y_{i\cdot\cdot\cdot} + \bar y_{\cdot j\cdot\cdot} + \bar y_{\cdot\cdot m\cdot} - \bar y_{\cdot\cdot\cdot\cdot})^2$
SSP error $E(C)$	$\sum_{i,j,m,\ell}(y_{ijm\ell} - \bar y_{ijm\cdot} - \bar y_{ij\cdot\ell} + \bar y_{ij\cdot\cdot})^2$
Total	$\sum(y_{ijm\ell} - \bar y_{\cdot\cdot\cdot\cdot})^2$

***F*-tests** (each $MS = SS/\mathrm{df}$):

- A: $F_A = MS_A/MS_{E(A)} \sim F_{a-1,\,(r-1)(a-1)}$, $H_{0A}: \mu_{1\cdot\cdot} = \cdots = \mu_{a\cdot\cdot}$.
- B: $F_B = MS_B/MS_{E(B)} \sim F_{b-1,\,a(r-1)(b-1)}$, $H_{0B}: \mu_{\cdot 1\cdot} = \cdots = \mu_{\cdot b\cdot}$.
- $A\times B$: $F_{AB} = MS_{AB}/MS_{E(B)} \sim F_{(a-1)(b-1),\,a(r-1)(b-1)}$, $H_{0AB}: \delta_{ij}^{AB} = 0\ \forall i,j$.
- C: $F_C = MS_C/MS_{E(C)} \sim F_{c-1,\,ab(r-1)(c-1)}$, $H_{0C}: \mu_{\cdot\cdot 1} = \cdots = \mu_{\cdot\cdot c}$.
- $A\times C$: $F_{AC} = MS_{AC}/MS_{E(C)} \sim F_{(a-1)(c-1),\,ab(r-1)(c-1)}$, $H_{0AC}: \delta_{im}^{AC} = 0\ \forall i,m$.
- $B\times C$: $F_{BC} = MS_{BC}/MS_{E(C)} \sim F_{(b-1)(c-1),\,ab(r-1)(c-1)}$, $H_{0BC}: \delta_{jm}^{BC} = 0\ \forall j,m$.
- $A\times B\times C$: $F_{ABC} = MS_{ABC}/MS_{E(C)} \sim F_{(a-1)(b-1)(c-1),\,ab(r-1)(c-1)}$, $H_{0ABC}: \delta_{ijm}^{ABC} = 0\ \forall i,j,m$.

Contrasts.

1. A marginal contrast $L_A = \sum_i p_i\mu_{i\cdot\cdot}$, $\sum_i p_i = 0$: $\hat L_A = \sum_i p_i\bar y_{i\cdot\cdot\cdot}$, $SE(\hat L_A) = \sqrt{MS_{E(A)}\sum_i p_i^2/(bcr)}$, $t = (\hat L_A - L_{A0})/SE(\hat L_A) \sim t_{(r-1)(a-1)}$. Pairwise A: $SE = \sqrt{2MS_{E(A)}/(bcr)}$.
2. B marginal contrast $L_B = \sum_j q_j\mu_{\cdot j\cdot}$, $\sum_j q_j = 0$: $\hat L_B = \sum_j q_j\bar y_{\cdot j\cdot\cdot}$, $SE(\hat L_B) = \sqrt{MS_{E(B)}\sum_j q_j^2/(acr)}$, $t \sim t_{a(r-1)(b-1)}$. Pairwise B: $SE = \sqrt{2MS_{E(B)}/(acr)}$.
3. C marginal contrast $L_C = \sum_m h_m\mu_{\cdot\cdot m}$, $\sum_m h_m = 0$: $\hat L_C = \sum_m h_m\bar y_{\cdot\cdot m\cdot}$, $SE(\hat L_C) = \sqrt{MS_{E(C)}\sum_m h_m^2/(abr)}$, $t \sim t_{ab(r-1)(c-1)}$. Pairwise C: $SE = \sqrt{2MS_{E(C)}/(abr)}$.
4. Simple B contrast within fixed $A = i$, marginal over C: $L_{B|i} = \sum_j q_j\mu_{ij\cdot}$, $\hat L_{B|i} = \sum_j q_j\bar y_{ij\cdot\cdot}$, $SE = \sqrt{MS_{E(B)}\sum_j q_j^2/(cr)}$, $t \sim t_{a(r-1)(b-1)}$.
5. Simple C contrast within fixed $A = i, B = j$: $L_{C|ij} = \sum_m h_m\mu_{ijm}$, $\hat L_{C|ij} = \sum_m h_m\bar y_{ijm\cdot}$, $SE = \sqrt{MS_{E(C)}\sum_m h_m^2/r}$, $t \sim t_{ab(r-1)(c-1)}$.

Variance components. $\hat\sigma_e^2 = MS_{E(C)}$, $\hat\sigma_s^2 = (MS_{E(B)} - MS_{E(C)})/c$, $\hat\sigma_w^2 = (MS_{E(A)} - MS_{E(B)})/(bc)$, $\hat\sigma_\rho^2 = (MS_{\mathrm{Blocks}} - MS_{E(A)})/(abc)$. Tests: $MS_{\mathrm{Blocks}}/MS_{E(A)} \sim F_{r-1,\,(r-1)(a-1)}$ for $H_0: \sigma_\rho^2 = 0$; $MS_{E(A)}/MS_{E(B)} \sim F_{(r-1)(a-1),\,a(r-1)(b-1)}$ for $H_0: \sigma_w^2 = 0$; $MS_{E(B)}/MS_{E(C)} \sim F_{a(r-1)(b-1),\,ab(r-1)(c-1)}$ for $H_0: \sigma_s^2 = 0$. "'

7. Permutation Tests for the Split-plot (Case II) and Repeated-measures (Case III) Designs

▶ General idea Both models have a *two-stratum* error structure: each whole-plot unit (unit (i,k) in Case II, subject (i,k) in Case III) carries a shared random effect, so observations within a unit are correlated while different units are independent. Pooling all y_{ijk} and reshuffling freely destroys this covariance and is invalid. The permutation strategy is therefore: *reduce the data to the stratum where the tested effect lives, then permute only within that stratum so that exchangeability holds under the null.*

For every whole-plot unit (i,k) form two orthogonal reductions:

- **Unit mean** (whole-plot stratum), averaging over the within factor:

$$\bar y_{i\cdot k} = \frac{1}{m}\sum_j y_{ijk}, \qquad m = c\ \text{(II)},\ b\ \text{(III)}.$$

  Case III: $\bar y_{i\cdot k} = \mu + \alpha_i + l_{ik} + \bar e_{i\cdot k}$ (carries A only).
  Case II: $\bar y_{i\cdot k} = \mu_{i\cdot} + b_k + w_{ik} + \bar e_{i\cdot k}$ (carries A and the crossed block b_k).
- **Within-unit deviation profile** (subplot stratum):

$$d_{ijk} = y_{ijk} - \bar y_{i\cdot k}.$$

  Here the whole-plot random effects (l_{ik}, or $b_k + w_{ik}$) cancel, leaving only σ_e^2 noise; the profile carries the B main effect and the interaction.

Tests of A use the unit means; tests of B and $A\times B$ use the deviation profiles; the two strata are independent and are permuted separately.

Key design difference. In Case III subjects are *nested* in A, so whole-plot permutation reshuffles whole subjects *freely* across A. In Case II the block b_k is *crossed* with A (each block holds one unit per A level), so to cancel b_k the A labels must be permuted *within each block* (restricted permutation). Within-unit permutations are identical in both cases.

Statistic and *p*-value. For each effect use the ANOVA sum of squares (equivalently the pseudo-F on the matching error stratum) as the statistic T, but calibrate it by its permutation distribution rather than an F-table. With M Monte-Carlo permutations,

$$\hat p = \frac{1 + \sum_{m=1}^{M}\mathbb{I}\{T^{(m)} \ge T_{\mathrm{obs}}\}}{M+1}.$$

▶ Main effects

A main effect (between-unit).

$$H_0^A:\ \alpha_1 = \cdots = \alpha_a \quad (\text{equiv. } \mu_{1\cdot} = \cdots = \mu_{a\cdot}).$$

Reduction: collapse each unit to its scalar mean $\bar y_{i\cdot k}$. *Permutation:* Case III — permute the A labels of the an unit means *freely* ($(an)!/(n!)^a$ allocations); Case II — permute the A labels of the units *within each block* ($(a!)^r$ allocations). *Statistic:* $\mathrm{SS}_A = m_0\sum_i(\bar y_{i\cdot\cdot} - \bar y_{\cdot\cdot\cdot})^2$. *Error:* $\sigma_l^2 + \sigma_e^2/b$ (III) or whole-plot error σ_w^2 (II). *Exact*, regardless of B or interaction (both are averaged out).

B main effect (within-unit).

$$H_0^B:\ \beta_1 = \cdots = \beta_b \quad (\mu_{\cdot 1} = \cdots = \mu_{\cdot c}\ \text{in II}).$$

Reduction: within-unit deviations d_{ijk}. *Permutation:* permute the B labels *within each unit* independently. *Statistic:* $\mathrm{SS}_B = N_0\sum_j \bar d_{\cdot j\cdot}^2$.

- If additivity is assumed (no interaction): exact, but the tested null is the *joint* hypothesis $\{\beta_j \equiv 0$ and $(\alpha\beta)_{ij} \equiv 0\}$ (all profiles flat).
- If interaction may be present and the marginal B effect is wanted, use **Freedman–Lane residual permutation**: fit the reduced model containing the interaction, $\hat\delta_{ij} = \bar d_{ij\cdot} - \bar d_{\cdot j\cdot}$; form residuals $r_{ijk} = d_{ijk} - \hat\delta_{ij}$; permute r within each unit, rebuild $d_{ijk}^* = \hat\delta_{ij} + r_{ijk}^*$, recompute SS_B. (Asymptotically valid.)

▶ Interaction effect

$$H_0^{AB}:\ (\alpha\beta)_{ij} \equiv 0 \quad (\delta_{ij} \equiv 0\ \text{in II}),$$

i.e. all units share the same expected within-profile irrespective of their A level. *Reduction:* the within-unit deviation profile $\mathbf{d}_{ik} = (d_{i1k},\dots,d_{imk})$ (a vector per unit). *Permutation:* permute the A labels of the *whole profiles* — move the entire vector, not single entries. Case III: subjects free across A; Case II: profiles depend only on σ_e^2 and are independent, so they may be permuted freely across A (or, conservatively, within blocks). *Statistic:* $\mathrm{SS}_{AB} = n\sum_{i,j}(\bar d_{ij\cdot} - \bar d_{\cdot j\cdot})^2$. *Exact*, and *unaffected by the B main effect*, since a nonzero β_j is a common vertical shift of all profiles to which the column-centred SS_{AB} is invariant.

Note the symmetry: the A test and the $A\times B$ test permute the *same thing* (unit labels across A); they differ only in whether each unit is reduced to a mean (scalar) or a deviation profile (vector).

▶ Simple effects Fixing one factor at a level often collapses the two-stratum structure to a single stratum, giving exact tests.

B within $A = i_0$ (compare $\mu_{i_0 1},\dots,\mu_{i_0 m}$).

$$H_0:\ \mu_{i_0 1} = \cdots = \mu_{i_0 m}.$$

Reduction: take the subset $A = i_0$; use within-unit deviations. Conditioning on a single A level removes the interaction (there is only one profile), so the within-unit null is clean. *Permutation:* permute the B labels within each unit of the $A = i_0$ group. *Statistic:* $\mathrm{SS}_{B(i_0)} = n\sum_j(\bar d_{i_0 j\cdot})^2$. *Error:* σ_e^2 only. *Exact.*

A within $B = j_0$ (compare $\mu_{1j_0},\dots,\mu_{aj_0}$).

$$H_0:\ \mu_{1j_0} = \cdots = \mu_{aj_0}.$$

Reduction: take the subset $B = j_0$; each unit now contributes a single value y_{ij_0k}, so the design reduces to a one-way between-unit layout. *Permutation:* Case III — permute the A labels of these single values freely; Case II — the value still carries b_k, so permute the A labels *within each block*. *Statistic:* between-A sum of squares of the $y_{ij_0\cdot}$. *Error:* $\sigma_l^2 + \sigma_e^2$ (III) or $\sigma_w^2 + \sigma_e^2$ (II). *Exact* — and it avoids the Satterthwaite mixed-error approximation needed by the parametric test.

▶ Summary

Effect	Reduction	Permute	Scheme (II / III)
A main	unit mean (scalar)	A labels	within block / subjects free
B main	within-unit dev.	B labels	within unit / within subject
$A\times B$	dev. profile (vec.)	A labels	profiles free across A
B at $A{=}i_0$	subset, dev.	B labels	within unit
A at $B{=}j_0$	subset, single val.	A labels	within block / subjects free

Validity. All "exact" claims rely on exchangeability: between-unit tests need only i.i.d. units; within-unit tests additionally need compound symmetry / sphericity of the within-unit covariance (implied by the single σ_l^2 or σ_w^2). Under non-spherical within-unit covariance, within-unit label permutation is only approximately valid; use a multivariate or sphericity-free statistic on the profiles instead.